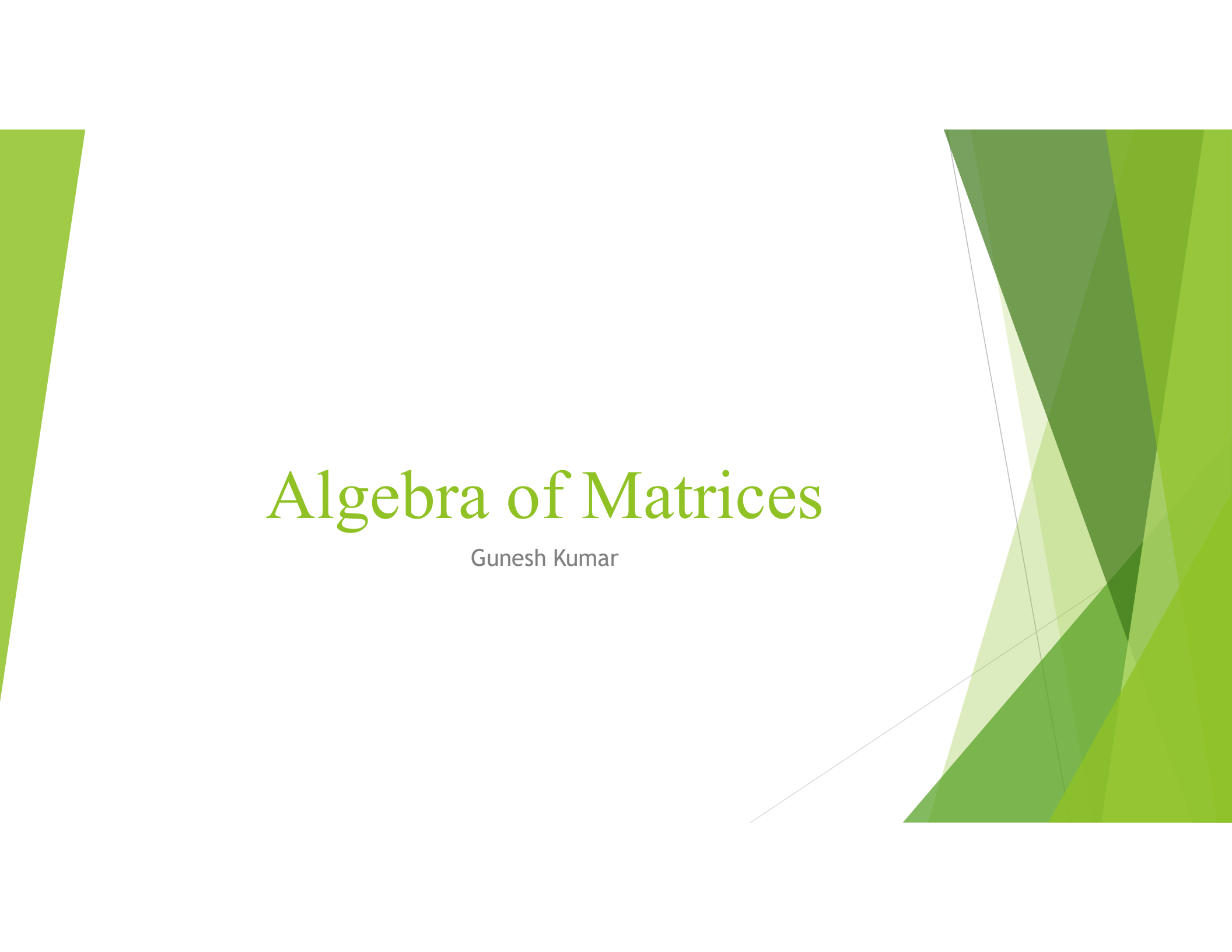


The background features abstract green geometric shapes. On the left, a solid green trapezoid points upwards. On the right, a complex arrangement of overlapping translucent green triangles and polygons creates a layered, crystalline effect. A thin, light gray line extends from the bottom left towards the right side of the page, passing behind the right-side geometric shapes.

Algebra of Matrices

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Definition: A square matrix A which $A^{k+1} = A$ (k being a positive integer), is called a periodic matrix.

If k is the least positive integer for which $A^{k+1} = A$, the A is said to be of period k .

If $k = 1$, so that $A^2 = A$, the A is called an idempotent matrix.

Definition: A square matrix A for which $A^p = 0$, (p being a positive integer), is called nilpotent.

If p is the least positive integer for which $A^p = 0$, then A is said to be nilpotent of nilpotency index p .

Definition: A square matrix A such that $A^2 = I$ is called an involutory matrix.



Q#1: If $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$, show that $A^2 - 4A - 5I = \begin{bmatrix} -2 & -2 & 2 \\ 0 & 0 & -2 \\ 0 & 0 & -2 \end{bmatrix}$

Solution:

$$A^2, \quad 4A, \quad 5I = ?$$

$$A^2 = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \times 1 + 2 \times 2 + 1 \times 2 & 1 \times 2 + 2 \times 1 + 1 \times 2 & 1 \times 1 + 2 \times 2 + 1 \times 1 \\ 2 \times 1 + 1 \times 2 + 2 \times 2 & 2 \times 2 + 1 \times 1 + 2 \times 2 & 2 \times 1 + 1 \times 2 + 2 \times 1 \\ 2 \times 1 + 2 \times 2 + 1 \times 2 & 2 \times 2 + 2 \times 1 + 1 \times 2 & 2 \times 1 + 2 \times 2 + 1 \times 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 + 4 + 2 & 2 + 2 + 2 & 1 + 4 + 1 \\ 2 + 2 + 4 & 4 + 1 + 4 & 2 + 2 + 2 \\ 2 + 4 + 2 & 4 + 2 + 2 & 2 + 4 + 1 \end{bmatrix} = \begin{bmatrix} 7 & 6 & 6 \\ 8 & 9 & 6 \\ 8 & 8 & 7 \end{bmatrix}$$

$$4A = 4 \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 8 & 4 \\ 8 & 4 & 8 \\ 8 & 8 & 4 \end{bmatrix}$$

$$5I = 5 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

$$A^2 - 4A - 5I = \begin{bmatrix} 7 & 6 & 6 \\ 8 & 9 & 6 \\ 8 & 8 & 7 \end{bmatrix} - \begin{bmatrix} 4 & 8 & 4 \\ 8 & 4 & 8 \\ 8 & 8 & 4 \end{bmatrix} - \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 7-4-5 & 6-8-0 & 6-4-0 \\ 8-8-0 & 9-4-5 & 6-8-0 \\ 8-8-0 & 8-8-0 & 7-4-5 \end{bmatrix} = \begin{bmatrix} -2 & -2 & 2 \\ 0 & 0 & -2 \\ 0 & 0 & -2 \end{bmatrix}$$

Proved

$$A^3 - 3A^2 - 7A - 3I = 0 \text{ (Home Work)}$$

Q#2: Show that $\begin{bmatrix} 0 & 1 & -1 \\ 4 & -3 & 4 \\ 3 & -3 & 4 \end{bmatrix}$ is involutory.

Solution:

For involutory $A^2 = I$

$$\begin{aligned} \text{So, } A^2 &= \begin{bmatrix} 0 & 1 & -1 \\ 4 & -3 & 4 \\ 3 & -3 & 4 \end{bmatrix} \begin{bmatrix} 0 & 1 & -1 \\ 4 & -3 & 4 \\ 3 & -3 & 4 \end{bmatrix} \\ &= \begin{bmatrix} 0+4-3 & 0-3+3 & 0+4-4 \\ 0-12+12 & 4+9-12 & -4-12+16 \\ 0-12+12 & 3+9-12 & -3-12+16 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{aligned}$$

$$A^2 = I$$

Proved

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Q#3: Show that the matrix $A = \begin{bmatrix} 1 & -2 & -6 \\ -3 & 2 & 9 \\ 2 & 0 & -3 \end{bmatrix}$ is periodic having period 2.

Solution:

For periodic $A^{p+1} = A$

$$\begin{aligned} A^2 &= \begin{bmatrix} 1 & -2 & -6 \\ -3 & 2 & 9 \\ 2 & 0 & -3 \end{bmatrix} \begin{bmatrix} 1 & -2 & -6 \\ -3 & 2 & 9 \\ 2 & 0 & -3 \end{bmatrix} \\ &= \begin{bmatrix} 1 + 6 - 12 & -2 - 4 + 0 & -6 - 18 + 18 \\ -3 - 6 + 18 & 6 + 4 + 0 & 18 + 18 - 27 \\ 2 + 0 - 6 & -4 + 0 + 0 & -12 + 0 + 9 \end{bmatrix} = \begin{bmatrix} -5 & -6 & -6 \\ 9 & 10 & 9 \\ -4 & -4 & -3 \end{bmatrix} \end{aligned}$$

$$A^3 = \begin{bmatrix} -5 & -6 & -6 \\ 9 & 10 & 9 \\ -4 & -4 & -3 \end{bmatrix} \begin{bmatrix} 1 & -2 & -6 \\ -3 & 2 & 9 \\ 2 & 0 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} -5 + 18 - 12 & 10 - 12 + 0 & 30 - 54 + 18 \\ 9 - 30 + 18 & -18 + 20 + 0 & -54 + 90 - 27 \\ -4 + 12 - 6 & 8 - 8 + 0 & 24 - 36 + 9 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -2 & -6 \\ -3 & 2 & 9 \\ 2 & 0 & -3 \end{bmatrix}$$

Here $A^3 = A$, so this is periodic.

$A^{2+1} = A$ the matrix having period is 2. Proved

Q#4: Show that $\begin{bmatrix} 1 & -3 & -4 \\ -1 & 3 & 4 \\ 1 & -3 & -4 \end{bmatrix}$ is nilpotent. What is the nilpotency index.

Sol: Let $A = \begin{bmatrix} 1 & -3 & -4 \\ -1 & 3 & 4 \\ 1 & -3 & -4 \end{bmatrix}$

$$A^2 = \begin{bmatrix} 1 & -3 & -4 \\ -1 & 3 & 4 \\ 1 & -3 & -4 \end{bmatrix} \begin{bmatrix} 1 & -3 & -4 \\ -1 & 3 & 4 \\ 1 & -3 & -4 \end{bmatrix}$$

$$= \begin{bmatrix} 1 + 3 - 4 & -3 - 9 + 12 & -4 - 12 + 16 \\ -1 - 3 + 4 & 3 + 9 - 12 & 4 + 12 - 16 \\ 1 + 3 - 4 & -3 - 9 + 12 & -4 - 12 + 16 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$
$$A^2 = 0$$

This A is the nilpotent and the order of nilpotency is 2.